

ADPKD-ECS: what we asked, and what we found

A short summary for a general reader. This was a laptop research program on cannabis-related questions in ADPKD (the inherited kidney-cyst disease), using only public data. It asked two separate questions. It is **not** advice to start or stop cannabis, CBD, or prescription drugs.

Study 1 — Is the cannabis receptor gene easy to see in the kidney cells that matter?

IDEA. Some papers suggest CB1 (the gene *CNR1*) is active in the kidney tubes that fail in ADPKD. Before testing “is it higher in disease?”, we locked a simple check: in publicly shared single-cell data, is *CNR1* even detectable in those tube cells at a pre-set floor (at least 1% of cells, and enough cells counted)?

WHAT WE DID. We froze the cell labels **without** using *CNR1* as a marker, then counted how often the gene showed up in 23,172 proximal-tubule nuclei from dataset GSE185948.

OUTCOME. It showed up in **51 cells (0.22%)**. That is below the 1% floor we had already written down. We stopped. We did **not** test whether it is higher in ADPKD than in healthy kidney, and we did **not** start computer drug-docking. This does **not** mean the receptor is absent from the kidney. It means this particular public map was too sparse to support the next claims.

Write-up: <https://github.com/cauld/adpkd-ecs/blob/main/research/07-scribe.md>

Preregistration: <https://osf.io/7g3tn/>

Study 2 — What can we actually say about cannabis/CBD with ADPKD and tolvaptan (Jynarque)?

IDEA. People with ADPKD sometimes use cannabis or CBD for pain. Kidney guidelines already say to ask about cannabis. Informal notes had overstated CBD as a proven problem with Jynarque. We locked a map in advance: plant cannabis, CBD, synthetic cannabinoids, and CB1 *drugs* must stay separate; CBD + Jynarque is “unstudied” unless a real pair study or a named label interaction shows up.

WHAT WE DID. We searched a frozen ClinicalTrials.gov query, quoted the 2025 KDIGO guideline, and read the FDA labels for Jynarque and Epidiolex (prescription CBD). We did not download patient gene-count files for this study.

OUTCOME. No matching ADPKD + cannabis/cannabinoid **trial** in that frozen search. KDIGO: ask about cannabis; **not recommended** as treatment (no shown benefit; possible kidney-injury risk from contamination/synthetics). The two labels **do not name each other**. Jynarque is broken down almost entirely by CYP3A and has a boxed liver warning. Epidiolex did not change a midazolam (CYP3A) probe; it can raise some other oral drugs. CBD + tolvaptan stays **unstudied** — not proven safe, not labeled as contraindicated. Analog results (midazolam, everolimus, grapefruit) are **not** a study of that pair.

Write-up: <https://github.com/cauld/adpkd-ecs/blob/main/studies/cannabis-safety/REPORT.md>

Preregistration: <https://osf.io/t6rzu/>

The whole program

There is no journal article. The two write-ups above are the public reports. Both studies live in one repository:

Repository: <https://github.com/cauld/adpkd-ecs>

Not medical advice. Do not use this summary to start, stop, or change cannabis, CBD, Jynarque, Epidiolex, or any other treatment. Study 1 does not show that CB1 is absent from the kidney. Study 2 does not show that CBD is safe or contraindicated with tolvaptan.

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